OENG1209: Engineering Science

Week 8:

Electric Charges & Electric Field

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Today's Topics

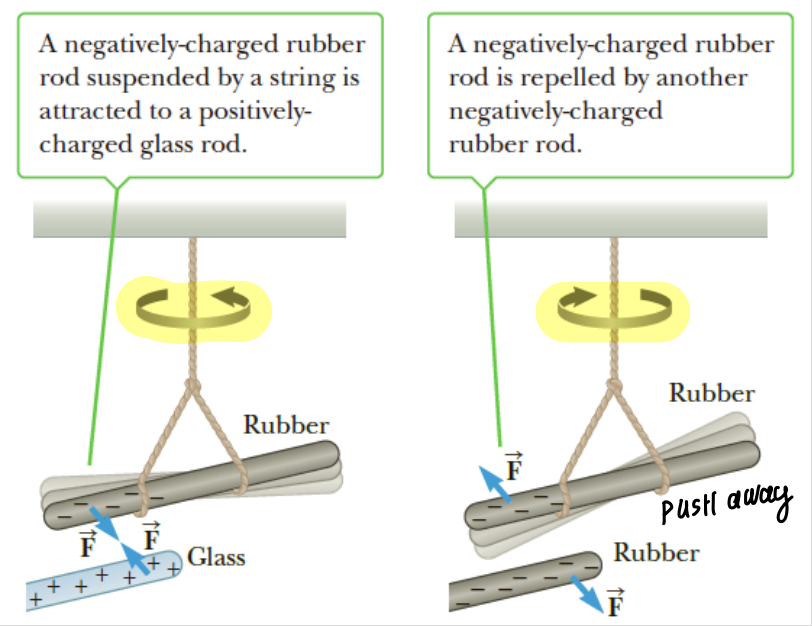
- Electric Charges
- Coulomb's Law
- Electric Fields
- Electric Potential Energy
- Potential Energy of Point Charges

1. Electric Charges

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Charges
Introduction

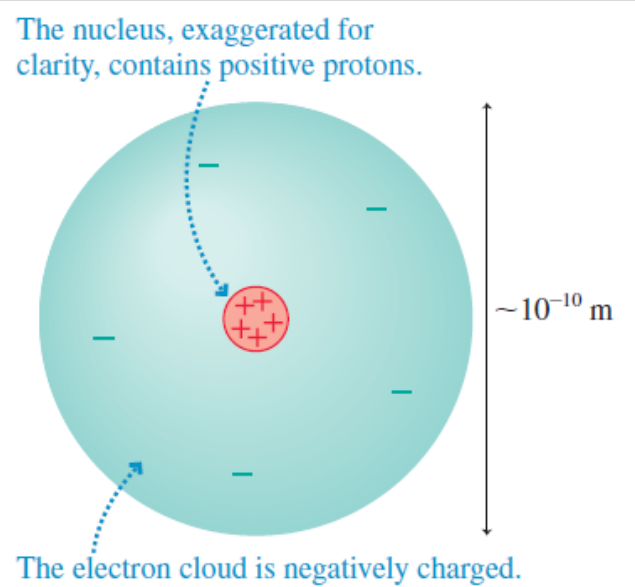
An experimental setup → the electrical force between two charged objects



- Two kinds or electric charge: **POSITIVE** & **NEGATIVE**
- **Like charges repel; unlike charges attract**

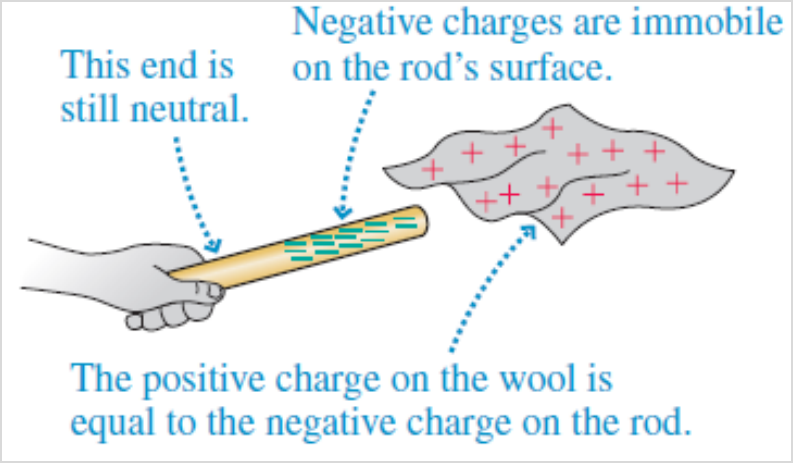
Charges
Atoms

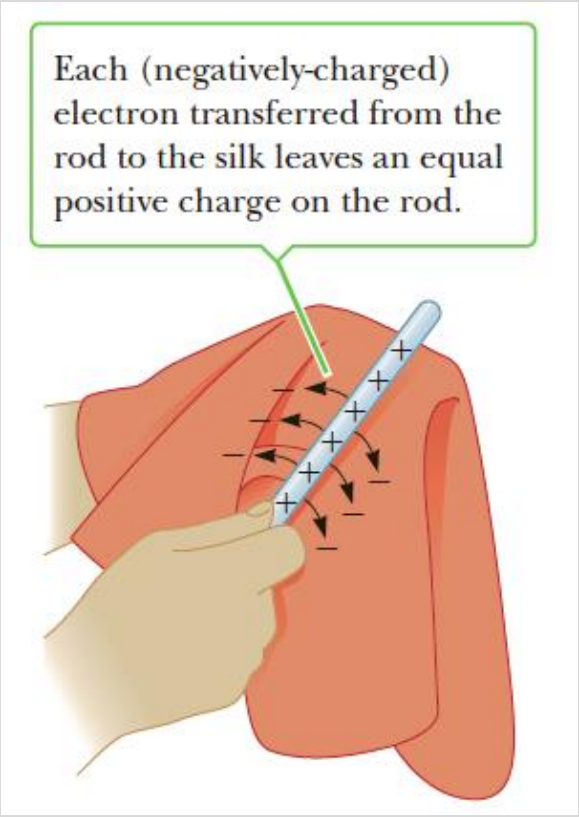
Consider an atom



Particle	Mass (kg)	Charge
Proton	1.67×10^{-27}	$+e$
Electron	9.11×10^{-31}	$-e$

Charges
Charging





2. Coulomb's Law

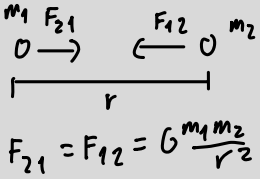
Quantifying the electrostatic forces

2 charges are at rest

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Coulomb's Law
Formula

analogous to Newton's law of gravity



$$F_{1\,on\,2} = F_{2\,on\,1} = \frac{K\,|q_1||q_2|}{r^2}$$

q_1 and q_2 are charges of the particles,

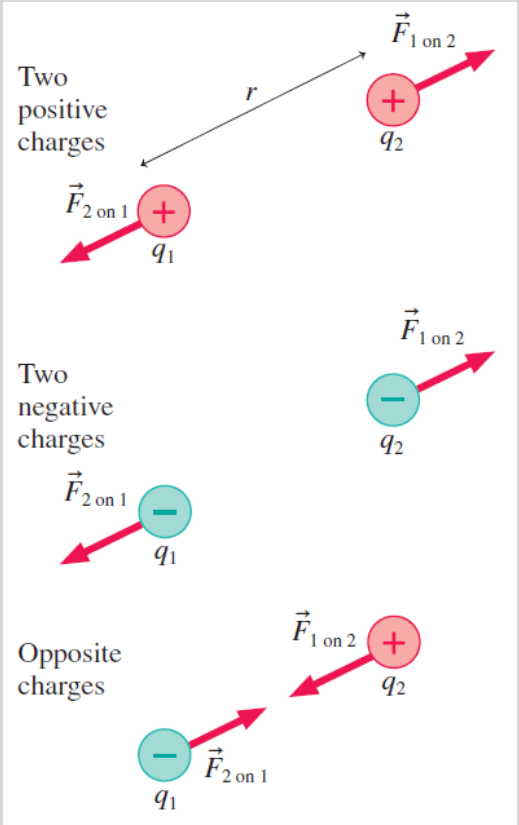
measured in Coulomb (C).

K is the electrostatic constant $= 8.99 \times 10^9 N m^2/C^2$

r is the distance measured between

the 2 particles in meters (m).

Remarks:



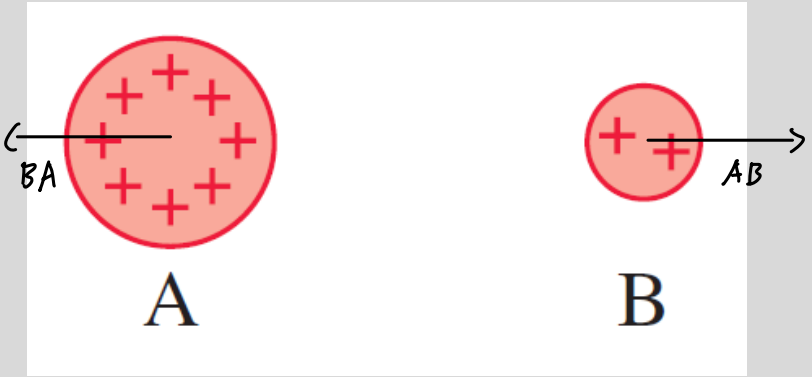
Coulomb's Law
Examples

Example: Charged spheres A and B exert repulsive forces on each other. Given that $q_A = 4q_B$, which statement is true?

a. $F_{\text{A on B}} > F_{\text{B on A}}$

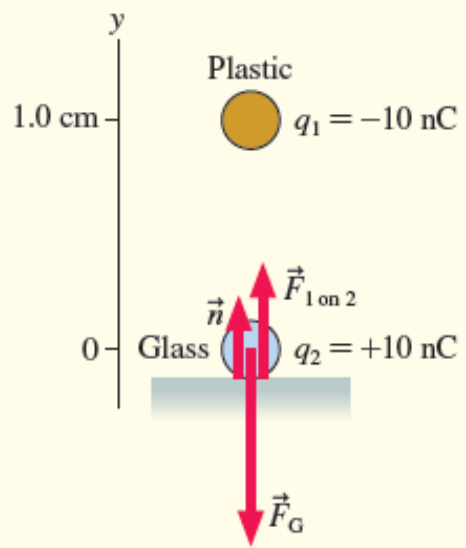
b. $F_{\text{A on B}} = F_{\text{B on A}}$ newton's 3rd law

c. $F_{\text{A on B}} < F_{\text{B on A}}$



Coulomb's Law
Examples

Example: A small plastic sphere charged to -10 nC is held 1.0 cm above a small glass bead at rest on a table. The bead has a mass of 15 mg and a charge of +10 nC. Will the glass bead "leap up" to the plastic sphere?



Electrostatic force:

$$F_{1on2} = k\frac{|q_1||q_2|}{r^2} = 9.00\times10^9\ \text{N m}^2/\text{C}^2\ \frac{(10\times10^{-9}\text{C})(10\times10^{-9}\text{C})}{(0.01\text{m})^2}$$

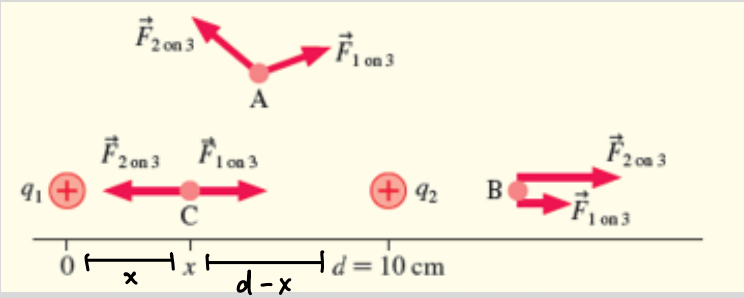
$$= 9\times10^{-3}\ \text{N}$$

Weight: $mg = 15\times10^{-6}\text{kg}\,(9.8\text{m/s}^2) = 1.5\times10^{-9}\ \text{N}$

∴ The glass will lift up

Coulomb's Law
Examples

Example: Where (other than at infinity) could a third charge q_3 be placed so as to experience no net force? Given $q_2 = 3q_1$



$F_{1-3} = F_{2-3}$

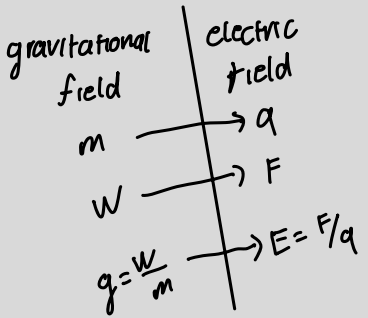
$F_{1-3} = k \dfrac{|q_1|\,|q_3|}{x^2}$

$F_{2-3} = k \dfrac{|q_2|\,|q_3|}{(d-x)^2}$

$\Rightarrow \dfrac{q_1}{x^2} = \dfrac{q_2 \nearrow 3q_1}{(d-x)^2} \Rightarrow (d-x)^2 = 3x^2,\ d = 10\text{ cm}$

$\Rightarrow x_1 = 3.66\text{ cm},\ x_2 = -13.66\text{ cm}$

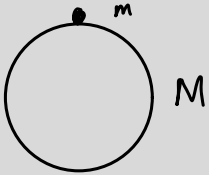
3. The Electric Field



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The Electric Field
Introduction

Recall the gravitational field



$$F = G\frac{Mm}{R^2} = gm \rightarrow W \Rightarrow g = \frac{W}{m}$$

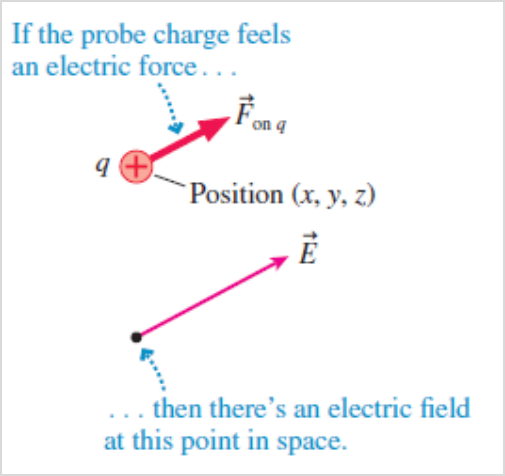
The space around a charge is altered by it to create the electric field $\vec{E}$,

$$\vec{E}(x,y,z) = \frac{\vec{F}_{on\,q}\ at\ (x,y,z)}{q}$$

The magnitude of vector $\vec{E}$ is called the

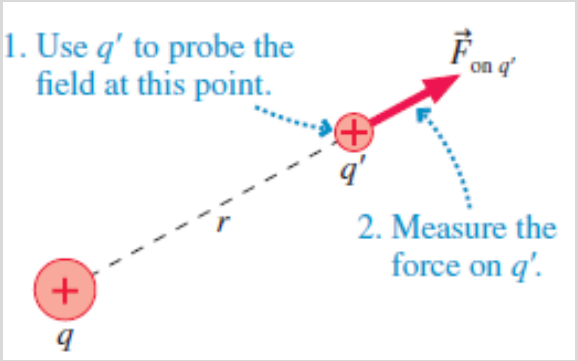
electric field strength.

The unit of $\vec{E} \rightarrow N/C$



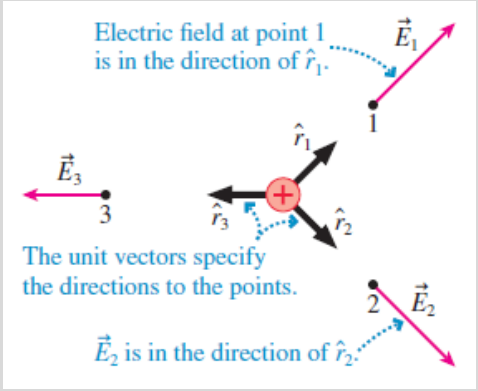
The Electric Field
Definition

Effect of the electric field on a point charge



$$\vec{F}_{q \to q'} = \vec{E}_{q'}$$

Using the unit vector $\hat{r}$



$$\vec{F}_3 = \vec{E}_{q3} = (E_{q3})\,\hat{r}_3$$

unit vector

$$\vec{r} = |\vec{r}| \cdot \hat{r}$$

mag

$\hat{r}$ ← unit vector

$\hat{r} = 1$

The Electric Field

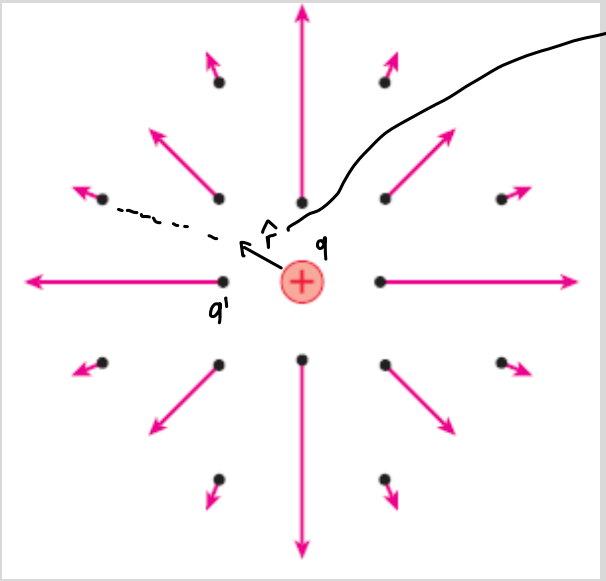
due to positive and negative charges

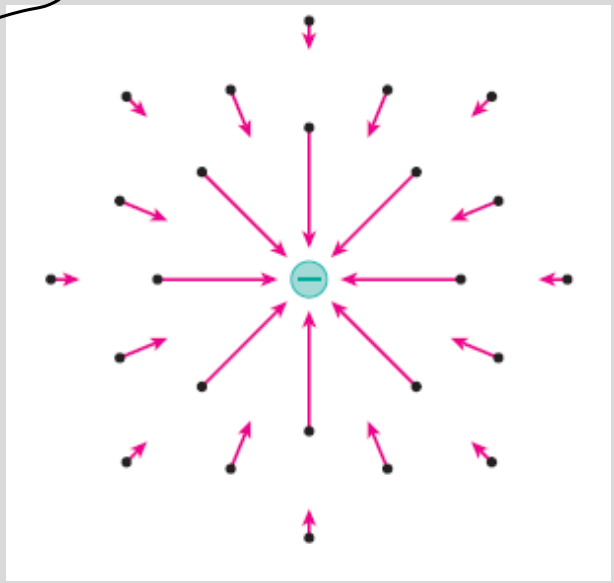
$$E = \frac{F}{|q'|} = \frac{1}{|q'|} k \frac{|q||q'|}{r^2} = k\,|q|/r^2 \qquad E \propto \frac{1}{r^2}$$

$$\vec{E} = \left(\frac{1}{4\pi\,\epsilon_0} \frac{|q|}{r^2} \hat{r} \right)$$

k

the direction

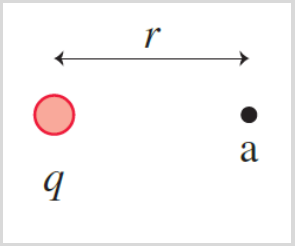




The Electric Field
Examples

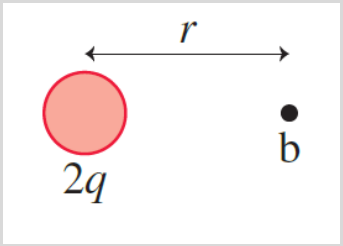
Example: Rank in order, from largest to smallest, the electric field strengths E_a to E_d at points a to d.

$E_a = k\frac{q}{r^2}$

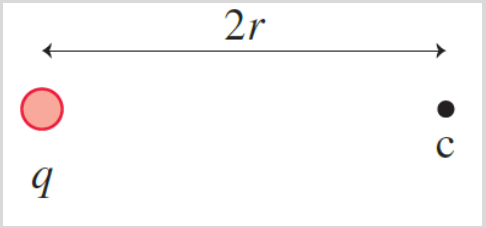


(a) ②

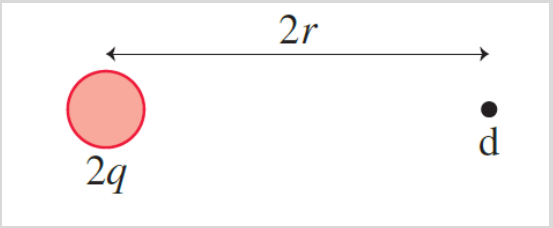
$E_b = k\frac{2q}{r^2}$



(b) ①



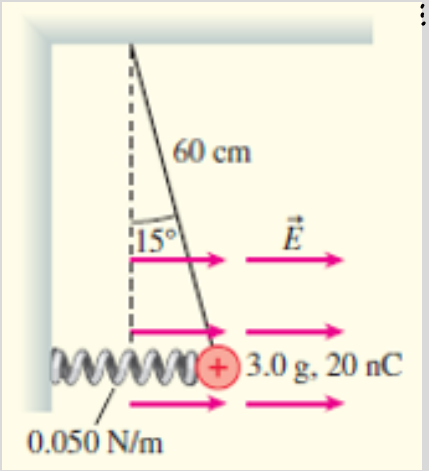
$E_c = k\,q/4r^2$ (c) ④

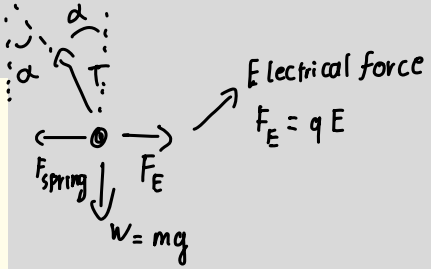


$E_d = k\frac{2q}{4r^2} = k\frac{q}{2r^2}$ (d) ③

The Electric Field
Examples

Example: A horizontal electric field causes the charged ball in Fig. to hang at a 15° angle, the spring is plastic, so it doesn't discharge the ball, and in its equilibrium position the spring extend only to the vertical dashed line. What is the electric field strength?





$F_s = k\Delta s$

$\Delta s = 60\,cm \sin 60$

$\uparrow \; T\cos\alpha - mg = 0 \Rightarrow T = \frac{mg}{\cos\alpha}$

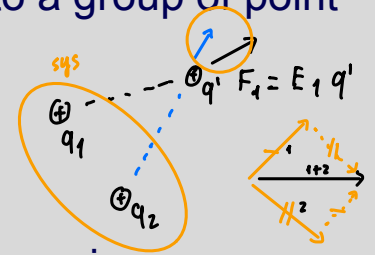
$qE - F_s - T\sin\alpha = 0$

$E = \frac{1}{q}\left(k\Delta s + mg\tan\alpha\right)$

The Electric Field
Multiple point charges

- **What happens if there is more than one charge?**
- Since Forces add as vectors, the net force on q due to a group of point charges is the vector sum:

$$\vec{F}_{on\ q} = \vec{F}_{1\ on\ q} + \vec{F}_{2\ on\ q} + \cdots$$



- Thus the net electric field due to a group of point charges is:

$$\vec{E}_{net} = \frac{\vec{F}_{1\ on\ q}}{q} + \frac{\vec{F}_{2\ on\ q}}{q} + \cdots = \vec{E}_1 + \vec{E}_2 + \cdots = \sum_i \vec{E}_i$$

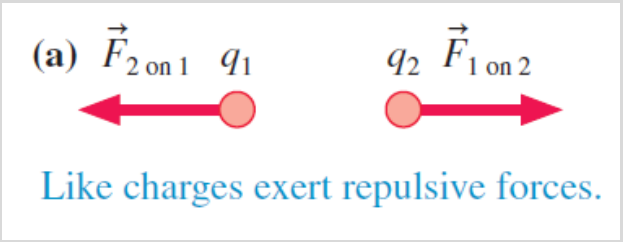
- This is also called the **principle of superposition**.

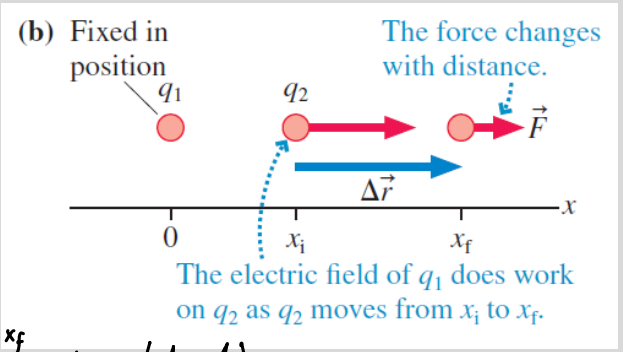
4. Potential Energy

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Potential Energy of Point Charges

Let us examine 2 charges q_1 and q_2 where q_1 had been "glued" down and is unable to move.





$$W = \int_{x_i}^{x_f} F\,dx = \int_{x_i}^{x_f} k\frac{q_1 q_2}{x^2}\,dx = k q_1 q_2 \int_{x_i}^{x_f} \frac{dx}{x^2} = -k q_1 q_2 \frac{1}{x}\Big|_{x_i}^{x_f} = -k q_1 q_2 \left(\frac{1}{x_f} - \frac{1}{x_i}\right)$$

Potential Energy of Point Charges

- If we defined our closed system to include both charges, we will have:

$$\Delta U_{elec} = U_f - U_i = -W_{elec} = \frac{Kq_1q_2}{x_f} - \frac{Kq_1q_2}{x_i}$$

- Thus, we have the **potential energy of the two-point-charge system** as:

$$U_{elec} = \frac{Kq_1q_2}{x}$$

$U = mgh \Rightarrow U = F \cdot x = k\frac{q_1q_2}{x^2} x = k\frac{q_1q_2}{x}$

- For a more general form of the equation, we will use r instead of x:

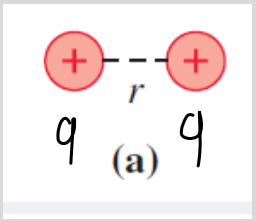
$$U_{elec} = \frac{Kq_1q_2}{r} = \frac{1}{4\pi\epsilon_0}\frac{q_1q_2}{r}$$

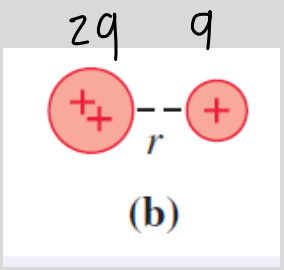
- Conservation of energy equation will also apply:

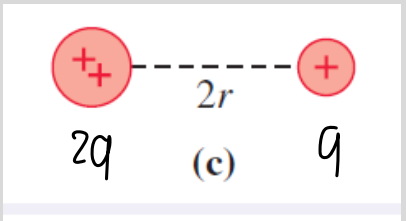
$$K_i + U_i = K_f + U_f$$

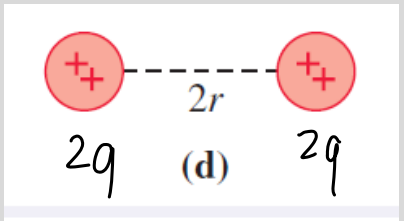
Potential Energy of Point Charges
Examples

Example: Rank in order, from largest to smallest, the potential energies U_a to U_d of these four pairs of charges. Each + symbol represents the same amount of charge.









$U_a = k\frac{qq}{r}$ $U_b = k\frac{2qq}{r}$ $U_c = k\frac{2q \cdot q}{2r}$ $U_d = k\frac{2q\,2q}{2r}$

$U_b = U_d > U_a = U_c$

5. The Electric Potential

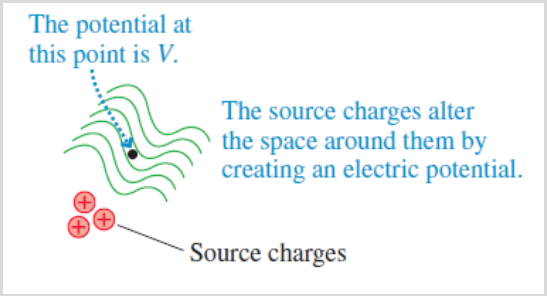
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The Electric Potential

- In analogy with the electric field, we will define the **electric potential *V*** which can be quantified as:

$$V = \frac{U_{q+sources}}{q}$$

- The electric potential, like the electric field, is a **property of the source charges**. And, like the electric field, the electric potential fills the space around the source charges.



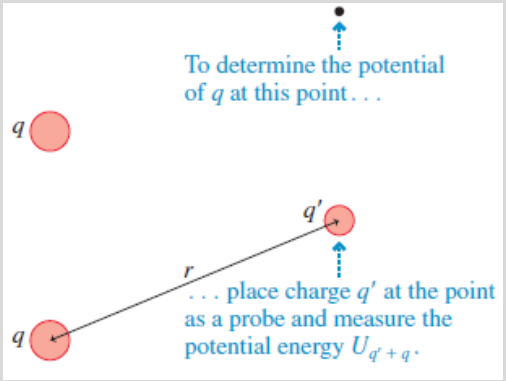


If charge q is in the potential, the electric potential energy is $U_{q\,+\,\text{sources}} = qV$.

The Electric Potential

Let q be the source charge and let a second charge q' probe the electric potential of q. **The potential energy** of the two point charges is:

$$U_{q'+q} = \frac{1}{4\pi\epsilon_0}\frac{qq'}{r}$$



- Hence, the electric potential of chare q is:

$$V = \frac{U_{q'+q}}{q'} = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$